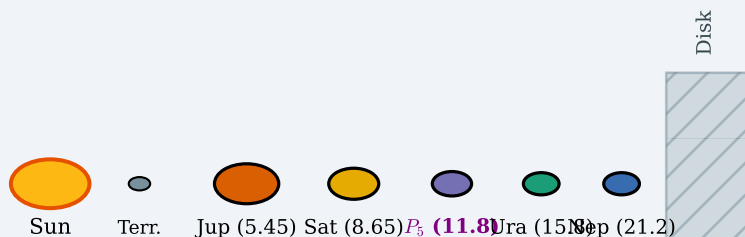


Dynamical Evolution of the Fifth Giant Planet Hypothesis (Nesvorny 2011)

Resolution of the Jumping-Jupiter Problem & Terrestrial Planet Coldness

Phase 1: Compact Primordial Multi-Resonance

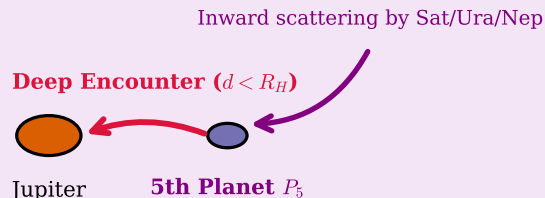
5 Giant Planets in Mean-Motion Resonant Chain (3:2, 3:2, 2:1, 3:2)



Compact configuration surrounded by massive outer planetesimal belt (22 – 30 AU)
 $35M_{\oplus}$

Phase 2: Planetesimal Scattering & Ice Giant Migration

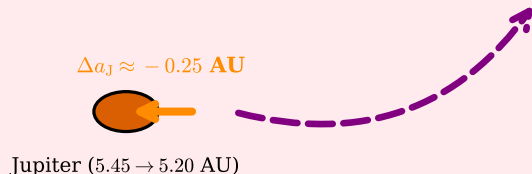
Resonance Crossing $\rightarrow$ Chaotic Planetary Orbit Crossing



Safronov parameter $\Theta_J \approx 37 \gg 1 \rightarrow$ Hyperbolic ejection kicks dominate

Phase 3: Hyperbolic Ejection & Jupiter Jump

Impulsive Jump: $\Delta a_J \approx -0.25$ AU, $\Delta(P_S/P_J) > 0.3$
Rogue Planet (Ejected into Interstellar Space)



Jump timescale $\tau_{\text{jump}} < 10^5$ yr $\ll$ Secular resonance sweeping timescale

Phase 4: Modern Outer Solar System Architecture

4 Giant Planets Survive; Terrestrial Planets Preserved Dynamically Cold

Canonical Architecture Reproduced ($R^2 \geq 0.99$)



Mars $e \leq 0.09$, Earth $e \leq 0.03$, Cold Kuiper Belt & Asteroids Preserved